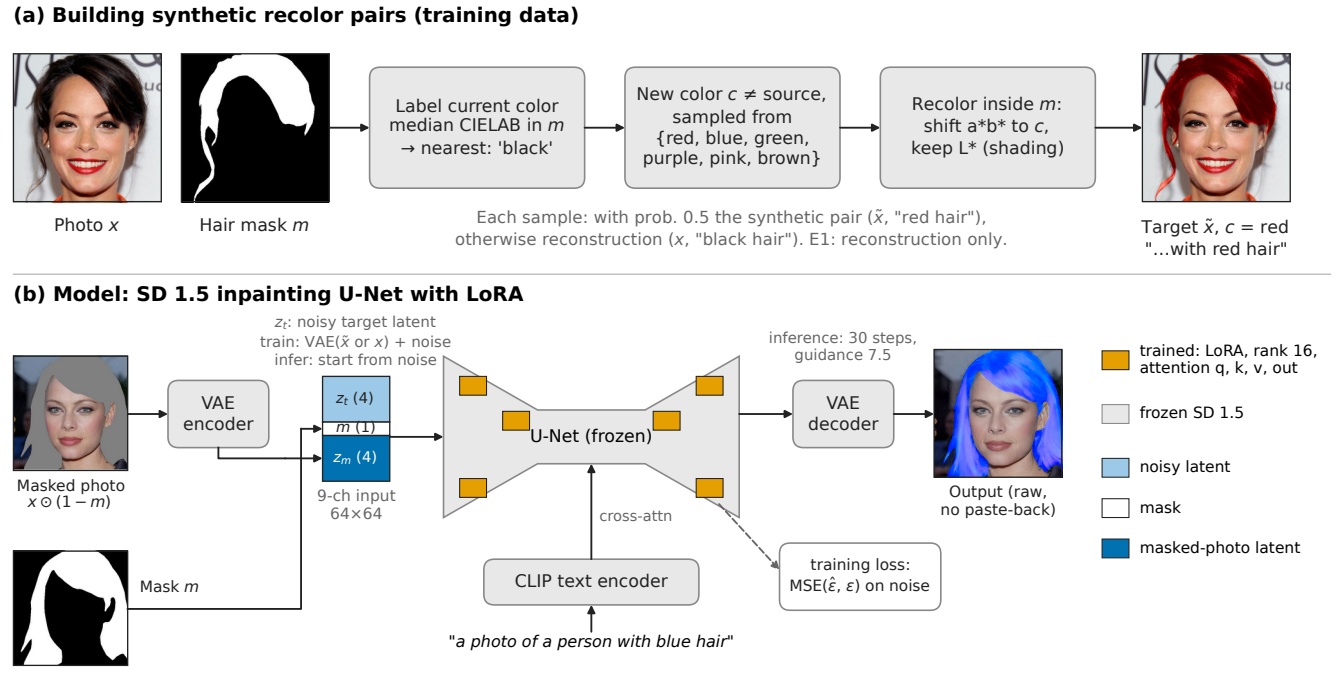

(a) Building synthetic recolor pairs (training data)
Photo x
Hair mask m
Label current color median CIELAB in m → nearest: 'black'
New color $c \neq$ source, sampled from {red, blue, green, purple, pink, brown}
Recolor inside m: shift a*b* to c, keep L* (shading)
Target $\tilde{x}$, c = red "...with red hair"
Each sample: with prob. 0.5 the synthetic pair ($\tilde{x}$, "red hair"), otherwise reconstruction (x, "black hair"). E1: reconstruction only.
(b) Model: SD 1.5 inpainting U-Net with LoRA
z_t: noisy target latent
train: VAE($\tilde{x}$ or x) + noise
infer: start from noise
Masked photo $x \odot (1-m)$
VAE encoder
z_t (4)
m (1)
z_m (4)
9-ch input 64×64
U-Net (frozen)
cross-attn
CLIP text encoder
"a photo of a person with blue hair"
Mask m
inference: 30 steps, guidance 7.5
VAE decoder
Output (raw, no paste-back)
training loss: MSE($\hat{\varepsilon}$, ε) on noise
trained: LoRA, rank 16, attention q, k, v, out
frozen SD 1.5
noisy latent
mask
masked-photo latent